

9.3 Finding and Interpreting the Mean, Median, Mode, and Range

Lesson Introduction

 Benchmark	MA.6.DP.1.2 Given a numerical data set within a real-world context, find and interpret mean, median, mode, and range.		 Learning Targets	- I can find the mean, median, mode, and range of a given data set. - I can interpret the mean, median, mode, and range of a given data set.	
 Benchmark Coherence	Building On			Working Toward	
	MA.5.DP.1.2 Interpret numerical data with whole-number values, represented with tables or line plots, by determining the mean, mode, median, or range.			MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, taking into consideration the context and any outliers.	
 Essential Questions	- How can you find and interpret the mean, median, mode, and range of a given data set? - What does the mean/median/mode/range of a data set tell you? - How does context impact the interpretation of mean/median/mode/range of a data set?				
 Lesson Narrative	In this lesson, students will explore and find the mean, median, mode, and range of a given data set. Students will interpret their findings to describe the data sets. Students will work with partners to explore the concepts of mean, median, and mode.				
 Component Summary	9.3.1 Warm-Up (~5 min)	Compare lengths of lines (<i>SE p. 50</i>)		9.3.4 Guided Instruction (10-15 min)	Find mean, median, and mode using definitions (<i>SE p. 52</i>)
	9.3.2 Exploration (5-10 min)	Discover how to redistribute groups of a square evenly (<i>SE p. 51</i>)		9.3.5 Try It (5 min)	Practice finding mean and median (<i>SE p. 54</i>)
	9.3.3 Collaboration (5-10 min)	Find the average (mean) of a given set of numbers (<i>SE p. 51</i>)		9.3.6 Exploration (5-10 min)	Practice finding mode and range (<i>SE p. 54</i>)
	9.3.7 Wrap-Up (~5 min)	Reflect on understanding of mean, median, mode, and range (<i>SE p. 56</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Mean - Median - Mode - Range - Measures of center - Measures of variability <u>Additional Terms:</u> N/A		(Optional) Calculator (Optional) Computer		Consider predetermining partners before the lesson to help with matching students appropriately.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may confuse mean and median or mode and median. - Students may think the mean or median must be one of the values in the data set. - Students may try to find the median before ordering the data set. - Students may forget to divide by the number of data values after finding their sum to determine the mean.	- Students may feel overwhelmed with finding four different measurements. Consider providing graphic organizers or chunking information. - Encourage students to write out all of the numbers in a data set in order from least to greatest every time they deal with the set to make sure they do not miss a number - Consider providing students with extra paper to solve and give more room to think. - Encourage students to show their work on every problem to avoid basic mistakes. - If time is an issue, then consider assigning 9.3.5 Try It for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share In 9.3.2 Exploration, students complete questions 1 and 2 independently. Then students discuss their responses and strategies with a partner. These discussions help students gain confidence and understanding.	MA.K12.MTR.7.1: Real-World Contexts Data throughout this lesson is presented through real-world contexts. Encourage students to connect the content of the lesson to their everyday lives by considering when summarizing data using the mean, median, mode, or range might be helpful. This will build investment in the mathematics.
 Differentiate Instruction	Notice the opportunity to use manipulatives and visuals during 9.3.2 Exploration. Visual entry points are available through the display of data in graphs and charts. Encouraging the use of manipulatives in this section will provide an additional kinesthetic entry point for learners to engage with the content. This is particularly helpful as students use manipulatives to create equal groups and compare this to the algorithm they may have used to find the mean. This connection deepens understanding of how to find the mean and can help students understand the context, particularly when determining the number needed to distribute evenly amongst groups. Students will see how certain data, such as animals or humans, cannot be split into fraction or decimals.	
Lesson Notes:		

9.3.1 Warm-Up: Stripes

Component Overview: In this Warm-Up, students will compare stripes on four groups of cubes. Students will discuss the similarities and differences between the sizes.



9.3.1 Warm-Up: Stripes

A picture of four sets of 16 cubes is shown.



For those who struggle with spatial reasoning, consider allowing students to cut out each colored strip to compare them more easily. Consider having students imagine how triangles could make up each line to also get a better visual on how long each is.

1. Which set of cubes has a set of stripes that covers the most area?

The set in the bottom left corner with the green stripes is the set with stripes that cover the most area.

2. Explain how you determined your answer.

Answers may vary. It has the most stripes, so they logically cover the most area.

3. Which set of cubes has a set of stripes that covers the least area?

The set in the top right corner with the blue stripes is the set with stripes that cover the least area.

4. Explain how you determined your answer.

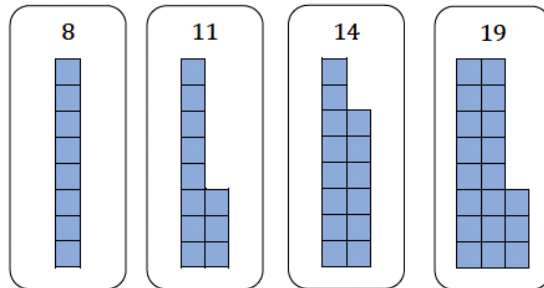
Answers may vary. If you put the blue stripes together, they will still be shorter than the one red stripe.

9.3.2 Exploration: Sharing Squares

Component Overview: In this Exploration, students evenly distribute squares into groupings. Students translate this into working with adding and dividing numbers to find the mean.



9.3.2 Exploration: Sharing Squares

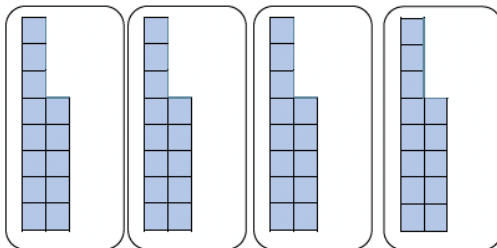


Consider providing square manipulatives for students to help understand the best way to distribute the squares evenly into four parts. This will provide a kinesthetic entry point.

1. Estimate the number of squares that will be in each group if they were distributed evenly between the four groups.

13

2. Use the diagram to redistribute the squares so that each of the four groups has the same number of squares.



3. Compare diagrams with your partner. Discuss how you each determined the number of squares that belong in each group. Summarize your discussion.

Answers will vary. Add all the blocks together and then divide by the number of cards, 4. I moved the blocks around from the columns with 14 and 19 so that the number in each became more even.



Students may struggle to explain how they determine the number of squares that belong in each group. To provide further guidance for discussion, ask students to consider the following questions:

- What did you do first to consider how to divide the squares?
- What does it mean to distribute evenly? Will each group have the same number? Why?
- How does the number of groups impact your answer?

9.3.3 Collaboration: Mean of Data

Component Overview: In this Collaboration, students will work together to find the mean of a data set using the given information. This page requires a heavy amount of reading, which may need to be broken down for struggling readers and ELLs. This section can be used as a guide for notetaking.



9.3.3 Collaboration: Mean of Data

By redistributing the squares in the diagram, the **mean** of the data set was found.



The **mean** is the arithmetic average of a set of numbers. It is a measure of central tendency.

Data in a numerical set is summarized using different measures that are interpreted to describe the data. One such measure used to describe quantitative data sets is the mean.

To calculate the mean, add all the values in the data set and then divide by the total number of data values.

1. A marine biologist is studying turtles and wants to calculate the average length of time a Red-Eared Slider, a popular breed of turtle, can remain underwater. The time, in minutes, that ten Red-Eared Sliders remain underwater are 75, 300, 225, 400, 250, 195, 320, 45, 165, 410.
 - a. To calculate the mean time spent underwater, start by finding the sum of the data values.
The sum is 2,385.
 - b. Then divide the sum by 10, because there are ten turtles.
 - c. In this sample, Red-Eared Slider turtles remain underwater for an average of 238.5 minutes.



Measures of center are numerical values used to describe the overall clustering of data in a set, or the overall central value of a set of data. The three most common measures of central tendency are the mean, median, and mode.



Turn and Talk

This section is intentionally scaffolded to build conceptual understanding of what mean is and why it is used. Have students engage in a Turn and Talk routine to restate the definition in their own words. This will ensure understanding over rote memorization.



Consider providing a graphic organizer to frame student thinking and help set up the proper way to find the mean of a set of numbers. Students may struggle with seeing numbers vertically and may need to view them horizontally.

9.3.4 Guided Instruction: Mean, Median, and Mode

Component Overview: In this Guided Instruction, students begin by filling in the definitions using terms from a word bank. Go through the definitions with the class to ensure students understand what each term means before moving on. Students then find the mean, median, and mode of a data set and describe the steps to find each of them.



9.3.4 Guided Instruction: Mean, Median and Mode

- For each definition, fill in the blank with one of the words from the bank.

median	mode
mean	range



The mean is the arithmetic average of a set of numbers. It is a measure of central tendency.



The range is the difference between the highest data value and the lowest data value in a data set. It is a measure of variation.



The mode is the value found most often in a set of numbers. There may be no, one, or more than one mode in the data set. It is a measure of central tendency.



The median is the middle of an ordered list of the values. If the list has an odd number of values, it is the middle value of that list. If the list has an even number of values, it is the average of the two middle values. It is a measure of central tendency.



Fill in the blanks for the glossary terms with the class. Begin by reading a definition to the class. Then have students talk with a partner to find and define the term in their own words. Review the correct answers as a class.



Allowing students to discuss each glossary term with a partner provides an auditory entry point to engage learners. Consider providing visual entry points, such as a histogram and data set with each term labeled, to engage visual learners.

9.3.4 Guided Instruction: Mean, Median, and Mode (continued)

The number of pets ten students in Tyler's class have is reflected in the table.

Number of Pets									
1	0	2	1	10	0	2	0	8	12

- Describe steps to find the median.
Order the numbers from least to greatest.
Find the middle number. If there is an even number of entries in the data set, find the average of the two middle numbers.
- The median number of pets for the ten students is 1.5.
- The mode of the data set is 0.
- Write an explanation of what the mode of the data set means for the context.
Answers will vary. The most common response is zero pets.
- Describe the steps needed to find the average number of pets owned by the ten students.
Find the sum of all the values in the data set. Then divide the sum by the number of values in the data set, 10.
- The mean number of pets for the ten students is 3.6.
- Complete the statement about the mean of the data set.
 "Each student in Tyler's class has about 3 or 4 pets."



Students may struggle to describe the steps to find the median. Provide support through questioning:

- What is the first thing you need to do to the data set to begin to find the median? Why?
- After the numbers are in order, how can you find the median? Is the number of data points significant? Why?

9.3.5 Try It: Mean, Median, and Mode

Component Overview: Students work independently or in pairs to interpret data in a table. Students find the mean and median of the data set and describe what they mean in the context of the question. Circulate as students work to complete this section and make note of any misconceptions to address either with individual students or as a class.



9.3.5 Try It: Mean, Median, and Mode

1. Nine adults boarding a cruise ship at Port Canaveral in Florida were randomly surveyed on the total number of vacations they have taken in the last five years. Their totals are summarized below.

Number of Vacations Taken								
5	4	3	9	1	4	0	6	3

- a. What does the value of the median represent in the context?
Half of the vacationers have taken more than the median number of vacations in the last five years and half have taken fewer.
- b. What is the median number of vacations taken by the adults surveyed?
4
- c. Determine the mean of the data. Then, interpret the mean in terms of the context by writing a sentence to summarize. Round to the hundredths.
The mean is 3.89 vacations. The people surveyed have taken an average of 3.89 vacations in the last five years.



As an extension, consider allowing students to discuss how the answers might change if they asked people on the street or a more general location instead of those boarding a cruise ship.

9.3.6 Exploration: Understanding Mode and Range

Component Overview: In this Exploration, students will work with a partner to discover the purpose of the mode and range using the definitions from the 9.3.4 Guided Instruction.



9.3.6 Exploration: Understanding Mode and Range

Work with your partner to complete the following.

1. The ☐ mean ☐ median ☒ mode ☐ range is the value occurring most often in a set of data, which can be used to describe trends in data.

2. While shopping over the last month, Shep collected the price for a jar of pickles from eight stores he visited.

Price for a Jar of Pickles			
Store A	\$1.66	Store E	\$1.99
Store B	\$2.99	Store F	\$1.99
Store C	\$1.49	Store G	\$1.50
Store D	\$1.99	Store H	\$1.99

- a. What does a jar of pickles cost at most stores?
\$1.99
- b. This value is the ☐ median of the data set. ☒ mode of the data set.
- c. What is the most Shep will spend on a jar of pickles?
\$2.99
- d. What is the least Shep will spend on a jar of pickles?
\$1.49
- e. The range of a data set describes how spread out the data is. Determine the range in price for a jar of pickles.
 $\$2.99 - \$1.49 = \$1.50$



To provide an additional visual entry point, consider encouraging students to represent the data on a number line for questions 2 and 3. This will clearly display the spread of the data and allow students to visually see the range and mode.

9.3.6 Exploration: Understanding Mode and Range (continued)

3. Shep also began collecting the price of his favorite cereal to determine where he would get the best price.

Price for a Box of Cereal			
Store A	\$3.79	Store E	\$3.99
Store B	\$3.89	Store F	\$3.89
Store C	\$3.99	Store G	\$4.29
Store D	\$3.89	Store H	\$4.59

- The mode price of his favorite box of cereal is **\$3.89**.
 - What is the most Shep will spend on a box of cereal?
\$4.59
 - What is the least Shep will spend on a box of cereal?
\$3.79
 - The range for the price of a box of cereal is **\$0.80**.
 - The range describes the

☐ center
☒ **variability**

 of the data set.
4. With your partner, discuss whether the price for pickles or the price for cereal has more variability in pricing. Summarize the discussion.

The price for pickles has more variability as seen by pickles having the larger range of prices.



Measures of variability are numerical values that measure how much a data set varies from a central value.



The range is a **measure of variability**. By comparing the ranges for two or more data sets, the data set that has more variability can be determined by finding which has a larger **range**.



Think-Pair-Share

Implement a Think-Pair-Share routine before students complete question 4. This will give students a chance to consider what it means to have more variability and formalize their responses with a partner before writing. Circulate as students discuss and bring the class together to address any misconceptions that may arise. Students may not connect that a larger range indicates more variability. This is an important misconception to address with the class.

9.3.7 Wrap-Up: Reflection

Component Overview: In this Wrap-Up, students will reflect on their understanding of the lesson, what areas they excel in, and what areas they need more support and help with.



9.3.7 Wrap-Up: Reflection

1. Read each of the “I can” statements below. Check the boxes that best explain your understanding.

Answers will vary.

- ☐ I can find and interpret the median, mode, and range of a given data set.
- ☐ I understand how to find the mean, median, and mode of a data set.
- ☐ I need support on how to find the mean, median, and mode of a data set.
- ☐ I understand how to find the range of a data set.
- ☐ I need support on how to find the range of a data set.
- ☐ I understand how to interpret measures of central tendency.
- ☐ I need support on how to interpret measures of central tendency.



Consider using Google Forms to survey students to identify who needs the lesson to be retaught or reinforced.